Italia Rodriguez

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EDUCATION

The University of Texas at Austin — **Expected: May 2026**

B.S., Biomedical Engineering — *Austin, TX*

- **GPA: 3.7/4.0** ; Engineering Honors
- **Coursework:** Development and Analysis in BME Design, Biomedical Instrumentation, Engineering Biomaterials, Biomedical Imaging Modalities
- **Study Abroad**: Micro- and Nanotechnology for Sensors | May 2025 | Kyoto, Japan
- **Organizations:** Biomedical Engineering Society, Gamma Rho Lambda

WORK EXPERIENCE

Detekt Biomedical · Austin, TX — **February 2024 – Present**

CAD Technician / Lead Assembly Technician / QC and Assembly Technician

- Prototyped and iterated accessory components for Detekt's flagship lateral flow assay reader (RDS-500) using SolidWorks, enabling additional device orientations and expanding overall usability.
- Designed adapter components to support wider cartridge compatibility, resulting in deployment across international clients including Inito, PerkinElmer, RMIT, Hardy Diagnostics, and BioPanda (15+ custom products).
- Produced detailed manufacturing drawings and modified existing designs to meet client specifications and manufacturability constraints.
- Lead assembly and pre-assembly operations for a production line averaging ~250 devices per month, ensuring consistent throughput, quality, and on-time delivery.
- Supervise and train 3 technicians across assembly and pre-assembly workflows, improving process consistency and reducing build variability.
- Designed and implemented a custom screen-alignment jig that reduced assembly errors by ~20%, improving first-pass yield and overall product quality.
- Assembled, manually calibrated, and quality-checked biomedical readers prior to shipment, ensuring compliance with internal quality standards.
- Troubleshot mechanical and calibration issues during production and communicated feedback upstream to support design and process improvements.

PROJECTS

International Medical Device Make-a-Thon

- Collaborated on the design of a menstrual disc concept capable of detecting ovarian cancer biomarkers, integrating biomedical sensing principles with user-centered design.
- Evaluated design tradeoffs related to usability, feasibility, and manufacturability under time constraints; placed 7th internationally among biomedical engineering teams.

Wearable UV Sensor & Mobile App

- Contributed to the design and construction of a wearable UV-exposure tracking prototype.
- Supported BLE communication integration, usability testing, and early-stage mobile app interface design for real-time exposure alerts.

SKILLS & INTERESTS

- **Technical Skills:** SolidWorks, MATLAB, Python, Arduino, Microsoft Office
- **Manufacturing & Production:** Assembly Line Operations, Process Improvement, Jig & Fixture Design, Yield Improvement, Root-Cause Troubleshooting, Technician Training
- **Volunteering:** Inclusivity Workshops, Central Texas Food Bank Volunteer
- **Interests:** Puzzles; Exercise; Digital Art; Travel; Podcasts; Fabric Crafts